Exploring Voice Banking as an Alternative Augmentative Communication Strategy for Individuals with Dysphonia, Aphonia, and Dysarthria: A Scoping Review☆

[*]Serena Pu, [†]Abbey Sawyer, [‡]Carrie Levinson, [†]David Putrino, and [*]Diana N. Kirke, *†*New York, New York*

SUMMARY: Objectives. This scoping review sought to: (i) review studies involving people diagnosed with dysphonia, aphonia, and dysarthria who have used voice banking technology in a hospital or community setting, and (ii) understand the scope of research surrounding existing voice banking technology and software in the clinical setting.

Methods. This scoping review was conducted according to the preferred reporting items for systematic reviews and meta-analyses extension for scoping reviews. An electronic search of databases, including Ovid MEDLINE (R), Embase (Ovid), APA PsycINFO (Ovid), and CINAHL was conducted. Title, abstract, and full text screening were completed using Covidence (Veritas Health Innovation, Melbourne, Australia) by two reviewers.

Results. After deduplication, 1336 studies underwent title and abstract screening. Of these, the full texts of 65 studies were reviewed, and 23 were included. Three distinct topics were identified in the search: (i) message banking, in which discrete messages or phrases are recorded for later use, without the ability to create novel phrases outside of those previously recorded; (ii) voice banking, in which all the necessary phonemes in the target language are captured, with the goal of generating a personalized synthetic voice that can generate novel phrases; (iii) and voice conversion or voice reconstruction, in which pathologic voices or speech are "converted" into a personalized synthetic voices.

Discussion and conclusion. This scoping review summarized the evidence for voice banking technology use in people diagnosed with dysphonia and dysarthria and sought to understand the research landscape of existing types of voice banking technology and software use in a clinical setting. The included papers were highly heterogeneous in terms of the population and type (research vs clinical program vs review/other). There is a pressing need for the publication of clinical programs and models facilitating the adoption of voice banking, especially within populations affected by conditions such as amyotrophic lateral sclerosis and laryngectomy, to address existing gaps and foster broader implementation and accessibility.

Keywords: Voice banking–Message banking–Voice conversion–Dysphonia–Dysarthria–Aphonia.

Abbreviations: AAC, Alternative and augmentative communication–ALS, Amyotrophic lateral sclerosis–MND, Motor neuron disease–TTS, Text to speech.

INTRODUCTION

Alternative and augmentative communication (AAC) refers to all communication methods used to supplement or replace speech, typically in individuals who have temporary or permanent impairments in speech. It can range from simple gestures, facial expressions, and blinking, to writing, use of sign language, or pointing to letters, to speech generation devices.[1] One such type of AAC is voice banking.

Voice banking refers to recording a large inventory of speech designed to sample all co-articulated sounds in a language, which is used to create a synthetic voice to approximate the speaker's voice.[2] It is a practice that has thus far mainly been used in progressive neurological diseases, such as amyotrophic lateral sclerosis (ALS), in which patients develop communication difficulties like dysphonia and dysarthria. In qualitative studies of patients who use voice banking or other AAC, patients describe these technologies as helping them preserve their sense of identity and fighting back against their disease.[3] Outside of research in ALS, there is preliminary work studying voice banking in other clinical settings, including in patients who are treated with total laryngectomy for laryngeal cancer, who become aphonic after their surgery.[5] However, a perceived limitation of current voice banking methods is that most do not cater to people who have already demonstrated affected speech, for example, people with a congenital speech impairment or people who have experienced neurological trauma or very sudden disease progression.[4] These people may face challenges in achieving accurate voice recordings due to the inherent variations in their speech patterns. As a result, tailored voice banking solutions that accommodate diverse speech characteristics are crucial for ensuring accessibility and effectiveness for all users, regardless of their pre-existing conditions or unique speech profiles.

Studies examining the clinical implications and patient-reported outcomes of voice banking are essential for

Accepted for publication October 14, 2025.
☆ Funding: None to disclose.
From the *Department of Otolaryngology – Head and Neck Surgery, Icahn School of Medicine at Mount Sinai, New York, New York; †Department of Rehabilitation and Human Performance, Icahn School of Medicine at Mount Sinai, New York, New York; and the ‡Levy Library, Icahn School of Medicine at Mount Sinai.
Address correspondence and reprint requests to Serena Pu, 1 Gustave L. Levy Place, New York 10029, New York. E-mail: serena.pu@mountsinai.org
Journal of Voice, Vol xx, No xx, pp. xxx–xxx
0892-1997

https://doi.org/10.1016/j.jvoice.2025.10.018

understanding its potential benefits and limitations in addressing the diverse needs of individuals affected by dysphonia, dysarthria, and aphonia. While the idea of voice banking is emerging in mainstream technology (eg, Apple® and Microsoft® products), and disability space (eg, Model Talker, Vocal ID, Acapela, The Voice Keeper all in ALS), few studies have been undertaken to determine the impact of voice banking on clinical and patient-reported outcomes in people with dysphonia, dysarthria, and aphonia. As voice banking gains traction in both mainstream technology and disability spaces, it becomes increasingly pertinent to investigate its efficacy across different conditions to inform evidence-based interventions and optimize support for affected individuals.

While previous review articles have extensively covered AAC literature, there is minimal focus on voice banking as a specific subcategory of AAC. However, with the rapid advancement of artificial intelligence, the time investment required for voice recording, a historical barrier to voice banking adoption, may be mitigated. Consequently, there is a need to conduct a scoping review of the existing literature on voice banking, particularly focusing on its application and effectiveness for individuals already experiencing vocalization impairments, and to determine awareness of voice banking amongst the patient and clinical community.

PURPOSE

The primary objective of this scoping review was to understand the utilization of voice banking technology among individuals diagnosed with dysphonia across both hospital and community settings. Additionally, this review sought to evaluate the scope of research surrounding existing voice banking technology and software in the clinical setting.

Considering these objectives, the research questions were:

1. What is the extent and breadth of research conducted on the utilization of voice banking technology in both hospital and community settings for people diagnosed with dysphonia, aphonia, and dysarthria?
2. How comprehensive is our understanding of the existing voice banking technology and software utilized within clinical settings for managing dysphonia and dysarthria, and what gaps exist in the current literature regarding its implementation, efficacy, and user experiences?

METHODS

Methodology

A scoping review of the literature was performed to systematically map the research done in this field and identify gaps in knowledge. This study adhered to the Joanna Briggs Institute methodological framework for conducting scoping reviews.[6] This scoping review is reported according to the preferred reporting items for systematic reviews and meta-analyses extension for scoping reviews.[7]

Eligibility

Given the need for a scoping review to cover a broad focus, the Population, Concept, Context framework is recommended.[6] As such, this scoping review included studies that met the following:

Criteria

Participants.

People of any age who have or will have dysphonia, aphonia, or dysarthria.

Concept.

Use of voice banking technology.

Context.

In a hospital or community setting.

Information sources

This scoping review considered materials published worldwide across any time period. Experimental and quasi-experimental study designs, including randomized controlled trials, non-randomized controlled trials, before and after studies, and interrupted time-series studies, were considered, as were observational, case-control, cross-sectional studies, single case reports, and qualitative studies. Reviews, opinion pieces, magazine articles, and other educational materials were also considered. Studies published in a language other than English, conference abstracts (where a corresponding full text could not be found), and studies where the full text was not available were excluded.

Search strategy

A medical librarian (CL) led the development of the search strategy, which incorporated key concepts, relevant text-words, and index terms. The search was peer reviewed by another medical librarian using the Peer Review of Electronic Search Strategies checklist.[8] An example of the search strategy is available in Table 1. An electronic search of databases, including Ovid MEDLINE(R) (Medical Literature Analysis and Retrieval System Online), Embase (Ovid), American Psychological Association PsycINFO (Ovid), and CINAHL (Cumulative Index to Nursing and Allied Health Literature) was conducted. The search spanned from database inception to January 23rd, 2024, and the search was re-run to reflect new literature on February 7th, 2025. The full search strategy for all databases is included in Appendix A. In addition to the electronic searches of databases, reference lists of relevant studies were also screened.

For the remaining studies, titles and abstracts were screened by two researchers (SP, AS). Full texts were then reviewed by the same two researchers to identify studies for inclusion.

TABLE 1.
Search Strategy Example (Ovid MEDLINE)

#	Searches
Intervention	
1	Communication Aids for Disabled/
2	((voice or speech or messag*) adj3 (personali?e* or bank* or synthe?i* or conversion or prosthes?s)).mp.
3	1 or 2
Participants	
4	dysphonia/ or aphonia/ or dysarthria/
5	exp neurodegenerative diseases/
6	(dysphoni* or aphoni* or dysarthri* or neurodegenerati* or "amyotrophic lateral sclerosis" or als or gehrig* or "motor neuron disease" or "motor neurone disease" or mnd).mp.
7	((limit* or loss or lost) adj3 (speech or voice)).mp.
8	4 or 5 or 6 or 7
9	3 and 8

Study screening and selection

Title and abstract screening were completed using Covidence software (Veritas Health Innovation, Melbourne, Australia). Duplicates were automatically removed from further screening. Pending screening numbers, an initial pilot screen of ~50 records was completed to ensure concordance in application of the inclusion and exclusion criteria between reviewers (SP, AS). Both reviewers then worked independently and were blinded to each other's decisions until screening was complete.

The full text of potentially eligible studies was then retrieved and assessed for eligibility by the two researchers. Any disagreement between the two researchers was resolved via discussion. If consensus couldn't be reached, a third reviewer (DK) was planned to be engaged; however, this did not occur. The reasons for the exclusion of full-text studies that did not meet the inclusion criteria were recorded.

Data extraction and charting

Microsoft Excel® software was used to extract data on the general characterization of the studies and the other for the description of the outcomes of interest.

Data was extracted by two researchers (SP, AS). Disagreement was resolved through discussions or with a third researcher (DK).

The general characteristics of the studies included: (i) the author, (ii) year of publication, (iii) study design, (iv) study objective, (v) the population and sample size, (vi) setting and country of origin, (vii) the intervention performed (if applicable), (viii) sample attributes, (ix) outcomes, (x) instruments used and type of technology, (xi) results.

Outcomes of interest were:

- Access to and use of voice banking
- Technology type
- Patient-reported outcomes, including, but not limited to, quality of life, self-efficacy and mood

As this is a scoping review, a formal risk of bias assessment was not undertaken.[6]

RESULTS

After deduplication (n = 516), 1336 studies underwent title and abstract screening. Of these, the full texts of 65 studies were reviewed, and 22 were included. In addition to the formal search, one study was identified through hand searching on Google Scholar,[9] for a total of 23 included studies (Figure 1).

The paper types and population studied in each can be found in Table 2. The papers were published between 2010 and 2025. Of the studies that recruited participants from these populations, the sample size was small (range 1–18 participants), except survey studies, which had larger sample sizes (range 10–209 participants). Included studies were conducted in the United States (n = 10), United Kingdom (n = 5), Italy (n = 1), Poland (n = 1), Czech Republic (n = 1), China (n = 1), New Zealand (n = 1), Taiwan (n = 1) and Turkey (n = 1).

Although our search was designed around voice banking specifically, three distinct topics emerged in the search results: (i) message banking, in which discrete messages or phrases are recorded for later use, without the ability to create novel phrases outside of those previously recorded; (ii) voice banking, in which all the necessary phonemes in the target language are captured, with the goal of generating a personalized synthetic voice that can generate novel phrases; (iii) and voice conversion or voice reconstruction, in which pathologic (dysphonic or dysarthric) voices are "converted" into a personalized synthetic voices. Both voice banking and voice conversion aim to create a personalized synthetic voice for later use, but voice conversion specifically involves changing or extrapolating a dysphonic or dysarthric voice and generating a non-pathologic voice with all necessary phonemes. Therefore, the results will be discussed separately for each concept. The primary language used in these technologies was English, but other languages included Italian (n = 1), German (n = 1), Czech (n = 1), Polish (n = 1), Turkish (n = 1), and Mandarin (n = 1).

Message banking

Message banking was the topic of 3 papers in this review. A summary is provided in Table 3. Because the search strategy was not designed around message banking specifically, some studies in the message banking literature that were not captured in this review; notably, Costello and colleagues' work pioneering message banking in the pediatric population.[10–12] The papers included in this section are those that also mention voice banking.[2,13,14]

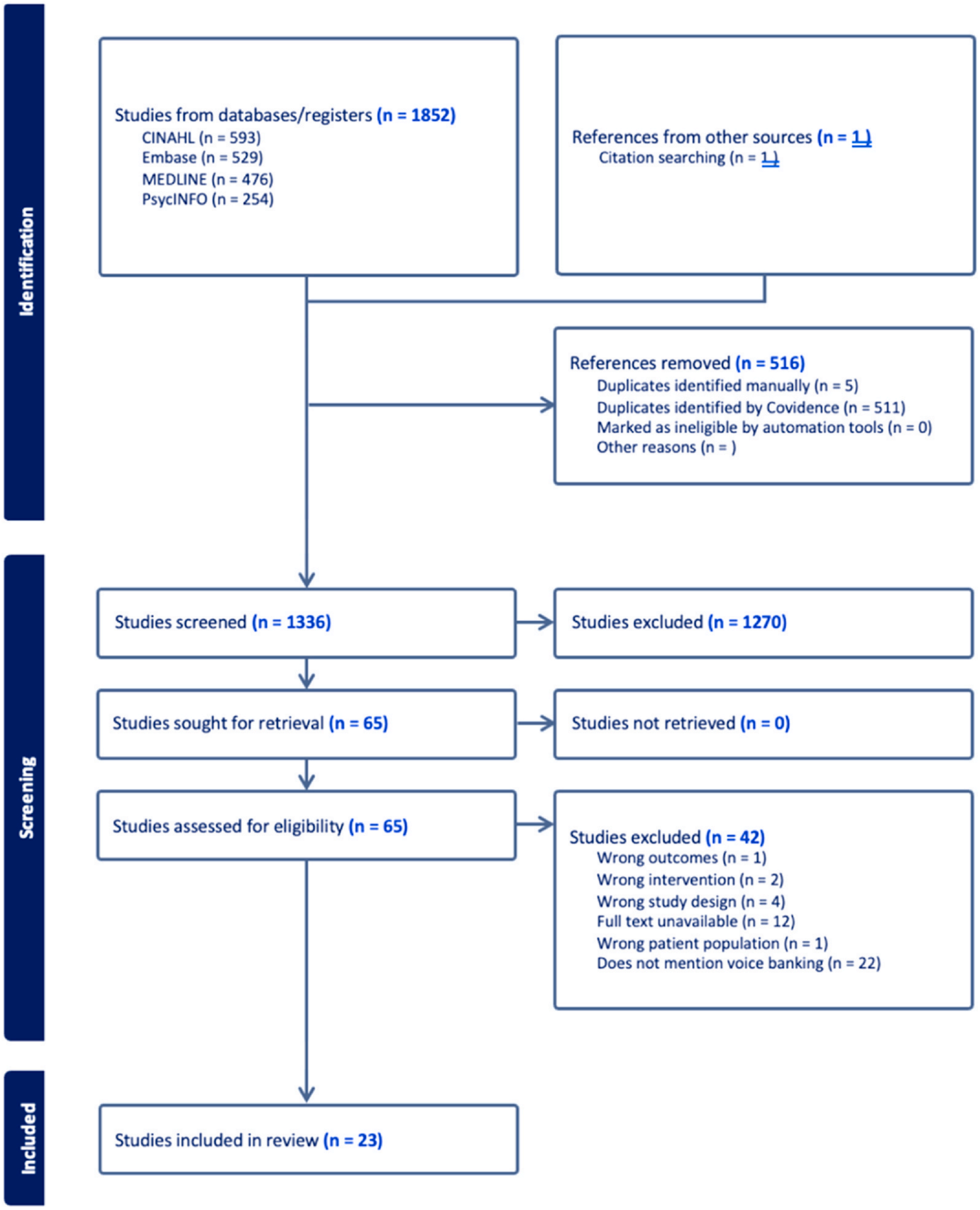

FIGURE 1. PRISMA-ScR flow diagram for the scoping review process. PRISMA-ScR, preferred reporting items for systematic reviews and meta-analyses extension for scoping reviews.

Voice banking/synthetic voice

There were 14 papers on the topic of voice banking and synthetic voice. Three papers were review articles: one mentioned voice banking in a discussion of future directions of AAC intervention for people living with motor neuron disease (MND).[15] The second paper mentioned voice and message banking in a discussion of high-tech AAC and specifically named Model Talker as a reliable system.[16] Both reviews referenced Costello and colleagues, among others.[11,12,17,18] The final paper was a narrative review on the effects of voice loss on self-concept and makes a case for personalized voice technology.[19] Voice banking was mentioned as an option for MND disease management in an educational article by Orrell.[20]

The remaining 10 papers will be summarized in Table 4. Total laryngectomy/laryngeal cancer was the main topic in 3 papers.[5,21,22] The remaining 7 papers discussed voice banking technology for patients with ALS (MND). English was the main language used in these voice banking studies, with Polish, Italian, and Czech as notable exceptions.[5,22,23] Two english dialects (New Zealand, Singaporean) were also included.[9,24]

The papers focused on a variety of facets surrounding voice banking and synthetic voices: the experience of and

TABLE 2.
Summary of Papers by Category

Paper type	Article	Population	
Research article (n = 13)	Cave and Bloch, 2021 Chen et al, 2023 Chu et al, 2023 Hyppa-Martin et al, 2023 Judge and Hayton, 2022 Mills et al, 2014 Oosthuizen et al, 2018 Peters et al, 2022 Regondi et al, 2025 Repova et al, 2021 Westley et al, 2019 Zhao et al, 2020 Zheng et al, 2023	ALS/MND (n = 12)	Arcuri, 2016 Cave and Bloch, 2021 Costello and Smith, 2021 Hanson et al, 2011 Hyppa-Martin et al, 2023 Judge and Hayton, 2022 Oosthuizen et al, 2018 Orrell, 2016 Peters et al, 2022 Regondi et al, 2025 Veaux et al, 2013 Zhao et al, 2020
Review (n = 3)	Fried-Oken et al, 2015 Hanson et al, 2011 Nathanson, 2017	Laryngectomy (n = 3)	Ahmad Khan et al, 2011 Repova et al, 2021 Szklanny and Lachowicz, 2022
Clinical or program summary (n = 2)_	Costello annd Smith, 2021 Veaux et al, 2013	Healthy voice donors (n = 2)	Chen et al, 2023 Westley et al, 2019
Case report (n = 2)	Ahmad Khan et al, 2011 Szklanny and Lachowicz, 2022	Pathologic voices (n = 1) Profoundly limited speech (n = 1)	Chu et al, 2023 Mills et al, 2014
Opinion piece (n = 1)	Turkmen and Karsligil, 2010	Dysarthric (n = 1)	Zheng et al, 2023
Educational article (n = 2)	Arcuri, 2016 Orrell, 2016	Dysphonic (n = 2)	Nathanson, 2017 Turkmen and Karsligil, 2010

Abbreviations: ALS, amyotrophic lateral sclerosis; MND, motor neuron disease.

motivation for voice banking,[3,9,24] the usage of synthetic voice,[1,25] and the perception/acceptability of synthetic voices.[9,21,22] One study provided an overview of their voice banking and voice reconstruction program.[26]

There were two studies evaluating the usage of voice banking technology by those to whom it was offered, ie a "service review."[1,27] They highlighted the importance of early intervention due to the time commitment required to consider their decision of voice banking, recording their voice, and adapt to using personalized synthetic voice devices, among other facets of disease management. One paper trained a combination of neural networks (HiFi-GAN: Generative Adversarial Network for Efficient and High Fidelity Speech Synthesis; and FastPitch) on recorded sentences obtained during voice banking to generate personalized synthetic voices.[23]

TABLE 3.
Summary of Papers on Message Banking

Authors	Paper type	Population	Summary
Costello and Smith (2021)	Summary of their experience	Pediatric patients at Boston children's hospital who have upcoming medical interventions that make speech difficult	Outlined principles for starting a message banking process, highlighted the possibility of "double dipping," ie combining the process of message banking and voice banking to generate a synthetic voice
Arcuri (2016)	Magazine article	n/a	Article on John Costello's work on message banking
Oosthuizen et al (2018)	Research	10 people with MND, their significant others, and speech-language pathologists	Limited awareness of message banking among all participant groups

Abbreviations: MND, motor neuron disease.

TABLE 4.
Summary of Voice Banking and Personalized Synthetic Voice Literature

Authors	Population	Language	Outcomes	Summary
Repova et al 2021	Patients undergoing total laryngectomy who were offered voice banking and personalized speech synthesis	Czech	Usage patterns of text to speech (TTS) technology	Greatest use of TTS technology in first postoperative week, which gradually decreased over time. Only 6/18 patients who banked their voices were active users of the technology. Subjective QoL measures improved in immediate postoperative period
Szklanny and Lachowicz 2022	Patient preparing for laryngectomy	Polish	Acceptance of synthetic speech; feasibility of their speech synthesis system	Synthetic voice created using statistical parametric speech synthesis system achieved acceptable quality by ratings of 25 expert listeners
Ahmad Khan et al 2011	Patient preparing for laryngectomy	English	Participants and family members were asked to judge the similarity of the synthetic voice to the preoperation (target) voice	The synthetic voice generated using model-based speech synthesis achieved acceptable similarity to target voice despite the low quality and small quantity (6–7 minutes only) of adaptation data
Cave and Bloch 2021	People with MND	English	Considerations for undertaking voice banking process	Many decisions about whether to undergo voice banking revolved around preservation of identity and perceived barriers to its use
Westley et al 2019	Healthy voice donors to ModelTalker speech synthesis program	New Zealand English	Perceptions and experiences with voice donation to ModelTalker	Voice donors found that the technology was easy to use and appreciated its benefits for those who use speech generating devices. The addition of more culturally relevant phrases would improve it for New Zealand users of speech-generating devices
Judge and Hayton 2022	Records of patients with MND (ALS) who were seen by neurological services in from 2018–19	English	Proportion of patients who were informed about voice banking, completed the voice banking process, using communication aids, and using synthetic voice regularly	Less than half of the clients had been informed about voice banking, one in four had completed the voice banking process, around half were using communication aids, and one in ten were using their personalized synthetic voice on a communication aid. The time taken to complete the process had a large variation.

TABLE 4 (*Continued*)

Authors	Population	Language	Outcomes	Summary
Peters et al 2022	People with MND who were surveyed about their experience	English	Current trends in the AAC use and service delivery experiences	Importance of early referral for AAC intervention, ongoing re-evaluation and treatment, involvement of communication partners and support for multimodal communication and adaptation to changing needs
Veaux et al 2011	People with MND	English	n/a	Overview of technical concepts behind ongoing project for voice banking and voice reconstruction program
Chen et al 2023	Healthy voice donors (Phase 1 and 2) Communication partners of people who use speech generating devices (Phase 3)	English (routinelyspoke Singapore Colloquial English)	The experience and perspectives of Singaporean voice donors in the voice banking process (Phase 1) Intelligibility and naturalness of banked synthetic voices created by Singaporean donors (Phase 2 and 3)	TTS English voices with a variety of accents are not readily available; thus, individuals who use SGDs may have synthetic voice output that does not always reflect theuniqueness of their linguistic community. Synthetic Singaporean-accented English voices can be produced using readily available web-based voice banking software (Model Talker). The average rating of ease of technology and software use was 8.43 (range: 7–9; SD: 0.79) on a 0 to 9 scale. Intelligibility was moderately high.
Regondi et al 2025	People with ALS with preserved voices (ALS-FRS score ≥1)	Italian	Technical quality and acoustic fidelity of personalized synthetic voices generated through HiFi-GAN model	10 ALS patients with voice preservation banked their voices, reading about 1000 sentences. A pretrained voice model was fine-tuned using neural networks with the target speaker's recorded voice. As compared to natural voices, the synthetic voices matched pitch, exceeded intensity, matched formant frequencies, and were perceived as similar (MCD and MOS). Overall, this model has potential for generating natural-sounding personalized synthetic voices.

Abbreviations: ALS, amyotrophic lateral sclerosis; ALS-FRS, amyotrophic lateral sclerosis functional rating scale; MCD, mel ceptral distance; MND, Motor neuron disease; MOS, mean opinion score; QoL, quality of life, SGD, speech-generating devices, SD, standard deviation.

Voice conversion

Similarly to message banking, voice conversion was not a specific search term, but rather a topic that emerged in the context of our voice banking searches. Therefore, some key voice conversion literature may be missing. There were 6 papers covering voice conversion.[4,28–32] The voice conversion process begins with dysphonic/dysarthric voices, and typically involves mapping features from the (pathologic/ dysphonic) source speaker into a voice-matched (non-dysphonic) speaker to create a synthetic voice with the acoustic features of the dysphonic speaker. Unlike voice banking, the generated synthetic voice does not directly replicate the original speaker's voice. Nonetheless, the goal is to create a comprehensible voice that can be identified to an individual; in other words, intelligible and personalized. Therefore, these voice conversion studies tested their own voice conversion technologies or techniques, and their outcomes were intelligibility of converted speech and listener preference. Mills and colleagues used speakers with profoundly limited speech motor control, but all other studies used pathologic voices or dysarthric speakers. Generally, the synthetic voices generated by voice conversion systems were able to improve intelligibility over non-converted (dysphonic or dysarthric) voices. These studies were conducted in various languages, including English, German, Turkish, and Mandarin. A summary of the studies is provided in Table 5.

DISCUSSION

This scoping review summarized the evidence for voice banking technology use in people with dysphonia, dysarthria, and aphonia and sought to understand the research landscape of existing types of voice banking technology and software use in a clinical setting. Whilst there were 23 papers included within this scoping review, the papers were highly heterogeneous in terms of the population and paper type (research vs clinical program vs review/other).

Voice banking methods have traditionally not catered to people who already demonstrate affected speech, for example, people with a congenital speech impairment or people who have experienced neurological trauma or rapid disease progression. This scoping review unveiled a related concept, voice conversion. When the input voice is already dysphonic and the generated synthetic voice "corrects" for the dysphonia, it may be more accurately characterized as voice conversion. One example of voice conversion technology is ModelTalker,[33] originating from the Nemours Speech Research Laboratory in Delaware, United States. This technology involves using synthesized voice-building to 'bank' voice for people whose natural voice is already impaired. The output is synthetic, at least in part, but representative of the natural voice. In one case report included in this scoping review,[21] voice recordings were made hastily by a female who was planned to undergo a laryngectomy. The recordings were made independently by reading passages of a novel on low-quality equipment with a sampling rate of 4 KHz. Two female average voice models were available, though these models had a different accent to the participant. Filters were able to be applied to clean and optimize the recording quality as much as possible, after which software was used to adapt the average female voice models to the target speaker material. This proof-of-concept case study provides a low-burden option for people who are unable to formally bank their voice in a controlled environment, and is a promising option for those who may be delayed in deciding to voice or message bank as well as those who are already experiencing early voice changes (eg hoarseness).[22]

The literature on voice conversion is promising for creating personalized synthetic voices for those with dysarthric speech, and for dysphonic or pathologic voices. The generated synthetic voice can retain recognizable acoustic features of their voice. While converted synthetic voices may be slightly more intelligible and preferred over dysphonic or dysarthric voices, they have not shown their converted synthetic voices to be non-inferior to typical, non-pathologic speakers.

Voice conversion technology may be further optimized with the rapid advances in artificial intelligence and deep learning. For example, speech neuroprostheses trained by artificial intelligence were augmented with preinjury voice recordings to replicate one patient's voice and speech, which were damaged due to neurological injury.[34] This approach highlights the growing role of machine learning in personalizing communication technologies, particularly when conventional speech is lost due to brain injury or neurodegenerative disease. Research involving brain-computer interfaces (BCIs) has also gained momentum in recent years, especially in the context of restoring communication in individuals with severe motor and speech impairments. BCIs are systems that translate brain signals into external outputs, allowing users to control devices or produce speech without relying on muscle movement.[35] Our review identified several recent studies that demonstrate the potential of intracortical speech neuroprostheses to enable real-time conversational communication for people with ALS who have severe dysarthria or are nearing complete loss of speech. For example, Card et al showed that direct decoding of neural activity from speech-related cortical areas could support intelligible and functionally useful communication in a person with ALS after a period of training.[36] Similarly, Angrick et al found that BCI-generated speech outputs achieved levels of intelligibility sufficient for untrained human listeners, representing a major milestone in the feasibility of BCIs as assistive communication tools.[37]

Some challenges surrounding voice banking technology include the patient's readiness to undertake the recording process. While this is best done early in the disease course, people living with ALS and their families often may not be ready to approach the idea of losing their speech, amongst the plethora of other medical advice coming their way. Patients who will become aphonic (eg undergoing total

TABLE 5.
Summary of Voice Conversion Literature

Authors	Language	Source Voice	Outcome	Technology	Summary
Chu et al 2023	German	Pathologic voices from Saarbrucken voice database	Intelligibility of converted speech (spectrogram analysis and word error rate; ratings by listeners)	n/a (they created E-DGAN: an encoder-decoder generative adversarial network)	The process of converting mel spectrogram of pathologic voices to normal, then synthesizing personalized speech with vocoder, resulted in increases in intelligibility and content similarity over non-converted voices
Hyppa-Martin et al 2023	English	Two people with ALS (mild dysarthria and moderate dysarthria) who completed voice banking	Synthetic speech intelligibility and naturalness as judged by listeners (n = 831), preferences of listeners about the synthetic speech and attitudes towards the synthetic speech	n/a (statistical parametric synthesis vs waveform concatenation techniques)	Synthetic voices generated by statistical parametric synthesis were preferred over voices generated by waveform concatenation, in terms of intelligibility, naturalness, and general preference. Synthetic voice generated from moderately dysarthric speech was preferred over the voice generated from mildly dysarthric speech
Mills et al 2014	English	Profoundly limited speech motor control (no intelligible vowels or consonants)	Intelligibility (normalized edit distance), speaker identification	VocaliD	The VocaliD conversion had minimal negative impact on intelligibility, but personalization was not effective for these talkers (vs for neurotypical talkers)
Turkmen et al 2010	Turkish	Dysphonic speech	Log spectral distances; subjective listening tests by 15 speakers (timbre, intelligibility, and naturalness	n/a	Created a speech reconstruction software using dysphonic speech; synthetic voice was preferred over dysphonic voice
Zhao et al 2020	English	ALS speakers with mild to severe dysarthria	Intelligibility and speech quality	n/a	Support for using voice conversion to assist speech communication for patients with ALS; robust to the full range of dysarthria severity, significant gain in intelligibility, and balance between quality and voice similarity
Zheng et al 2023	Mandarin	Ten mild-to-severe dysarthria patients and six normal speakers	Intelligibility (Google automatic speech recognition system, subjective listening test)	n/a (they created DVC 3.1, dysarthria voice conversion)	The experimental results showed that the DVC 3.1 system achieved higher speech intelligibility performance for listeners under free-talk testing conditions. Based on the results of this study, they suggest that the DVC 3.1 system can reduce the burden of dysarthric patients to record training data.

Abbreviation: ALS, amyotrophic lateral sclerosis.

laryngectomy) may place voice banking low on their list of priorities relative to their oncologic care. Readiness to accept AAC is a common hurdle, and as such, voice banking can be placed lower down on the priority list, by which time the opportunity to voice bank may be missed (or the quality impaired). Cave and colleagues sought to understand what people with ALS consider in the early stages following diagnosis (1–12 months) when deciding whether to bank their voice, what their expectations are, as well as the expectations of significant communication partners, through a series of semi-structured interviews.[3] Identity was the dominant theme in decision-making for people with ALS and their communication partners, with polarity in views around the impact of voice on identity itself.

Another barrier to voice banking lies in patient education and access. Judge and Hayton reviewed the records of patients with MND (ALS) who were seen by neurological services over the period of 2018–19, and less than half who were eligible to undergo voice banking had been informed about voice banking.[27] Healthcare providers should play an active role in encouraging patients to consider AAC options. In a 2022 study by Peters et al, even though most participants with MND had heard of voice/message banking, 62.1% of participants had not undertaken the voice banking process.[1] Access to technology may be a prohibitive factor. Synthetic voice or recorded messages require the use of some type of speech generating device (eg, smartphone, laptop, tablet, or purpose-built SGD). Participants in the study by Peters et al who did not have an SGD despite recommendations or had an SGD and did not use it cited concerns about support for learning and using the device, difficulty using the device, the device not working well, and high cost.[1]

One of the review's objectives was to understand the scope of research surrounding existing voice banking technology and software in the clinical setting. Whilst voice banking is anecdotally completed for clinical populations, with some organizations offering support for completion (eg, Team Gleason in collaboration with the Boston Children's Hospital) relatively few studies formally report on these clinical applications. This scoping review was limited to papers written in the English language. Consequently, important work in other languages may have been missed. The small number of papers included, along with the heterogeneity of sources, imposes constraints on the reviewers' ability to synthesize findings cohesively. However, this highlights priorities for future research and clinical care.

To address some of the barriers to voice banking, early counseling and repeated efforts to educate patients and their families about voice banking and other communication methods can help improve knowledge about and acceptance of AAC options. However, the highest priority should lie in finding or creating a user-friendly, accessible voice banking platform. Voice banking improves the quality of life for patients with voice loss and can help preserve a sense of identity.[3,5,19] Making such technology widely available to patients, whether they have a progressive neurologic disease or have another condition where they will lose their voice, can improve the quality of patient care and quality of life.

CONCLUSION

This scoping review summarized the evidence for voice banking technology use in people diagnosed with dysphonia and sought to understand the research landscape of existing types of voice banking technology and software use in a clinical setting. There is a pressing need for the publication of clinical programs and models facilitating the adoption of voice banking, especially within populations affected by conditions such as ALS and laryngectomy, to address existing gaps and foster broader awareness, accessibility, and implementation.

Registration

This protocol was registered on the Open Science Framework (https://osf.io/a2rm8/).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jvoice.2025.10.018.

References

1. Peters B, O'Brien K, Fried-Oken M. A recent survey of augmentative and alternative communication use and service delivery experiences of people with amyotrophic lateral sclerosis in the United States. *Disabil Rehabil Assist Technol.* 2022;101255937:1–14. https://doi.org/10.1080/17483107.2022.2149866.
2. Costello J, Smith M. The BCH message banking process™, voice banking, and double-dipping™. *AAC Augment Altern Commun.* 2021;37:241–250. https://doi.org/10.1080/07434618.2021.2021554.
3. Cave R, Bloch S. Voice banking for people living with motor neurone disease: views and expectations. *Int J Lang Commun Disord.* 2021;56:116–129. https://doi.org/10.1111/1460-6984.12588.
4. Mills T, Bunnell HT, Patel R. Towards personalized speech synthesis for augmentative and alternative communication. *AAC Augment Alternat Commun.* 2014;30:226–236. https://doi.org/10.3109/07434618.2014.924026.
5. Repova B, Zabrodsky M, Plzak J, Kalfert D, Matousek J, Betka J. Text-to-speech synthesis as an alternative communication means after total laryngectomy. *Biomed Pap Med Fac Univ Palacky Olomouc Czechoslovakia.* 2021;165:192–197. https://doi.org/10.5507/bp.2020.016.
6. Peters MDJ, Godfrey C, McInerney P, Munn Z, Tricco AC, Khalil H. Scoping reviews (2020). In: Aromataris E, Lockwood C, Porritt K, Pilla B, Jordan Z, eds. *JBI Manual for Evidence Synthesis.* JBI; 2024. https://doi.org/10.46658/JBIMES-24-01.https://synthesismanual.jbi.global.

7. Tricco AC, Lillie E, Zarin W, et al. PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. *Ann Intern Med.* 2018;169:467–473. https://doi.org/10.7326/m18-0850.
8. McGowan J, Sampson M, Salzwedel DM, Cogo E, Foerster V, Lefebvre C. PRESS peer review of electronic search strategies: 2015 guideline statement. *J Clin Epidemiol.* 2016;75:40–46.
9. Chen M, Hyppa-Martin J, Bunnell HT, et al. Voice banking to support individuals who use speech-generating devices: development and evaluation of Singaporean-accented english synthetic voices and a Singapore colloquial english recording inventory. *Augment Altern Commun.* 2023;39:208–218. https://doi.org/10.1080/07434618.2023.2181213.
10. Costello J. AAC intervention in the intensive care unit: the children's hospital Boston model. 2000/01/01 *Augment Altern Commun.* 2000;16:137–153. https://doi.org/10.1080/07434610012331279004.
11. Costello J. Last words, last connections. 12/15 *ASHA Leader.* 2009;14:8–11. https://doi.org/10.1044/leader.FTR2.14162009.8.
12. Santiago RCJ. AAC assessment and intervention in pediatric ICU/acute care: from referral through continuum of care. *Perspect Augment Altern Commun.* 2013;22:102–111.
13. Arcuri L. Giving patients a voice when they need one most. *Hospitals Health Netw.* 2016;90:19.
14. Oosthuizen I, Dada S, Bornman J, Koul R. Message banking: perceptions of persons with motor neuron disease, significant others and clinicians. *Int J Speech Lang Pathol.* 2018;20:756–765. https://doi.org/10.1080/17549507.2017.1356377.
15. Hanson EK, Yorkston KM, Britton D. Dysarthria in amyotrophic lateral sclerosis: a systematic review of characteristics, speech treatment and augmentative and alternative communication options. *J Med Speech Lang Pathol.* 2011;19:12–30.
16. Fried-Oken M, Mooney A, Peters B. Supporting communication for patients with neurodegenerative disease. *NeuroRehabilitation.* 2015;37:69–87. https://doi.org/10.3233/NRE-151241.
17. Costello J, Dimery K. *Preserving legacy: a guide to message banking in a patient's voice*; 2014.
18. Costello JM. *Message banking, voice banking and legacy messages*. Boston Children's Hospital, Boston, MA; 2014.
19. Nathanson E. Native voice, self-concept and the moral case for personalized voice technology. *Disabil Rehabil.* 2017;39:73–81. https://doi.org/10.3109/09638288.2016.1139193.
20. Orrell RW. Diagnosis and management of motor neurone disease. *Practitioner.* 2016;260:17–21.
21. Ahmad Khan Z, Green P, Creer S, Cunningham S. Reconstructing the voice of an individual following laryngectomy. *AAC Augment Altern Commun.* 2011;27:61–66. https://doi.org/10.3109/07434618.2010.545078.
22. Szklanny K, Lachowicz J. Implementing a statistical parametric speech synthesis system for a patient with laryngeal cancer. *Sensors.* 2022;22:3188–3206. https://doi.org/10.3390/s22093188.
23. Regondi S, Donvito G, Frontoni E, et al. Artificial intelligence empowered voice generation for amyotrophic lateral sclerosis patients. *Sci Rep.* 2025;15:1361. https://doi.org/10.1038/s41598-024-84728-y.
24. Westley M, Sutherland D, Bunnell HT. Voice banking to support people who use speech-generating devices: New Zealand voice donors' perspectives. *Perspect ASHA Specl Int Gr.* 2019;4:593–600. https://doi.org/10.1044/2019_PERS-SIG2-2018-0011.
25. Repova B, Romportl J, Betka J. Voice rehabilitation after total laryngectomy by TTS synthesis. *Otolaryngol Head Neck Surg.* 2015;153(1suppl 1):91. https://doi.org/10.1177/0194599815593290e.
26. Veaux C, Yamagishi J, King S, Colville S. Voice banking and voice reconstruction for MND patients. *Amyotrophic Lateral Sclerosis and Frontotemporal Degeneration.* 2013;14(suppl 2):79–80. https://doi.org/10.3109/21678421.2013.838414/031.
27. Judge S, Hayton N. Voice banking for individuals living with MND: a service review. *Technol Disabil.* 2022;34:113–122. https://doi.org/10.3233/TAD-210366.
28. Chu M, Yang M, Xu C, et al. E-DGAN: an encoder-decoder generative adversarial network based method for pathological to normal voice conversion. *IEEE J Biomed Health Inform.* 2023;27:2489–2500. https://doi.org/10.1109/JBHI.2023.3239551.
29. Hyppa-Martin J, Lilley J, Chen M, Friese J, Schmidt C, Bunnell HT. A large-scale comparison of two voice synthesis techniques on intelligibility, naturalness, preferences, and attitudes toward voices banked by individuals with amyotrophic lateral sclerosis. *Augment Altern Commun.* 2023;8504574:1–15. https://doi.org/10.1080/07434618.2023.2262032.
30. Turkmen HI, Karsligil ME. Dysphonic speech reconstruction: construction of a novel system for effective and efficient communication. *IEEE Eng Med Biol Magazine Quart Magazine Eng Med Biol Soc.* 2010;29:135–138. https://doi.org/10.1109/MEMB.2009.935725.
31. Zhao Y, Kuruvilla-Dugdale M, Song M. Voice conversion for persons with amyotrophic lateral sclerosis. *IEEE J Biomed Health Inform.* 2020;24:2942–2949. https://doi.org/10.1109/JBHI.2019.2961844.
32. Zheng W-Z, Han J-Y, Chen C-Y, Chang Y-J, Lai Y-H. Improving the efficiency of dysarthria voice conversion system based on data augmentation. *IEEE Trans Neural Syst Rehabil Eng.* 2023;31:4613–4623. https://doi.org/10.1109/TNSRE.2023.3331524.
33. Bunnell HT, Lilley J, Pennington C, Moyers B, Polikoff J. *The ModelTalker system*. Citeseer; 2010.
34. Anderer S, Hswen Y. Digital avatars and personalized voices—how ai is helping to restore speech to patients. *JAMA.* 2024;331:1259–1261. https://doi.org/10.1001/jama.2023.23556.
35. Patrick-Krueger KM, Burkhart I, Contreras-Vidal JL. The state of clinical trials of implantable brain–computer interfaces. 2025/01/01 *Nat Rev Bioeng.* 2025;3:50–67. https://doi.org/10.1038/s44222-024-00239-5.
36. Card NS, Wairagkar M, Iacobacci C, et al. An accurate and rapidly calibrating speech neuroprosthesis. *N Engl J Med.* 2024;391:609–618. https://doi.org/10.1056/NEJMoa2314132.
37. Angrick M, Luo S, Rabbani Q, et al. Online speech synthesis using a chronically implanted brain-computer interface in an individual with ALS. *Sci Rep.* 2024;14:9617. https://doi.org/10.1038/s41598-024-60277-2.